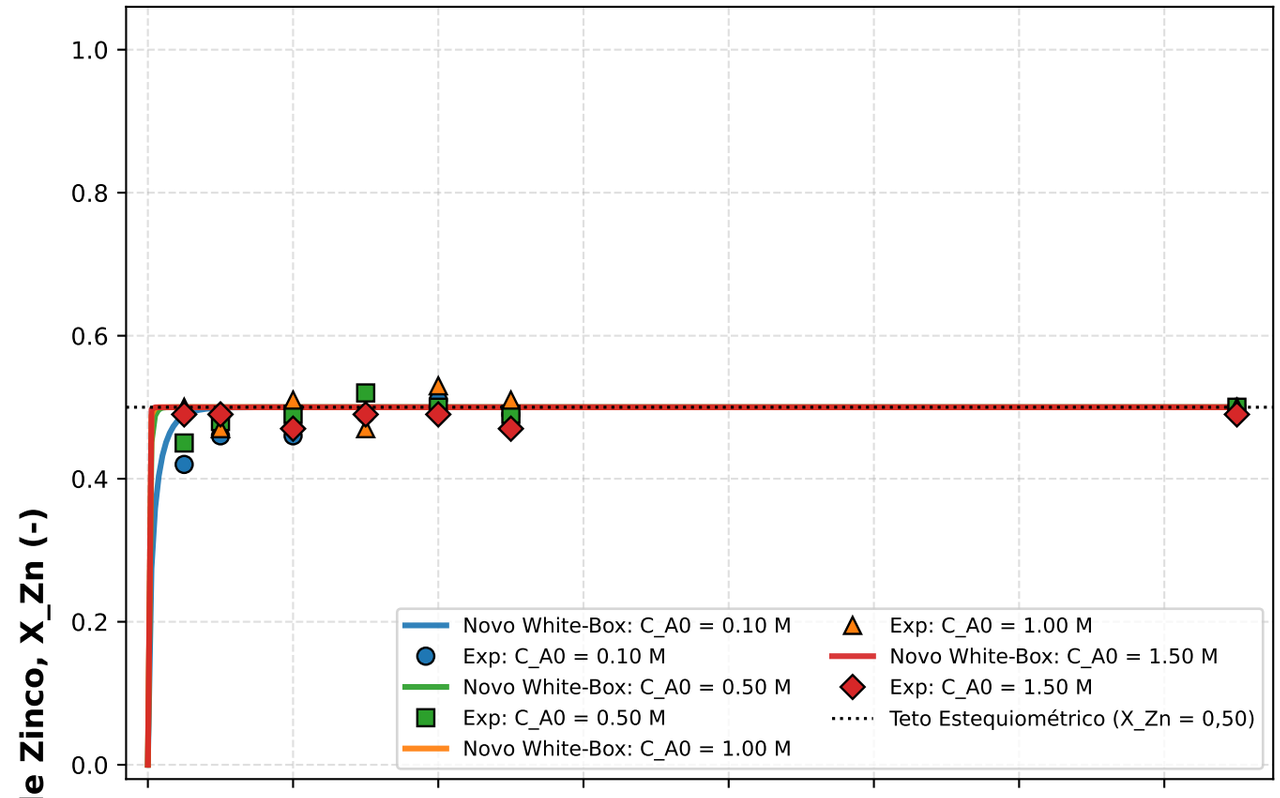
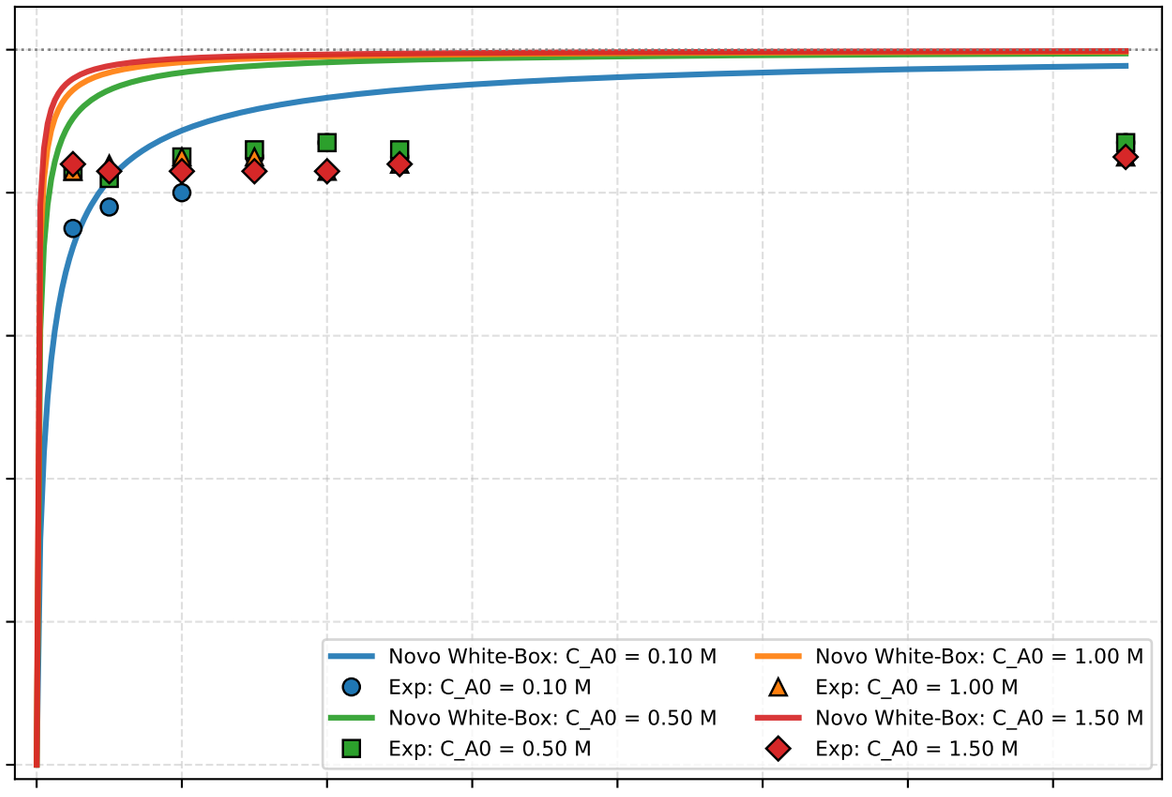
**Novo Baseline White-Box Puro Oficial (RRB ÷ D³, Sem Parâmetro α) vs. Dados Experimentais (t > 0)**
**Métricas Globais (112 Pontos Ativos, t > 0): R² = 0.8709 | RMSE = 7.02% | MAE = 5.16%**

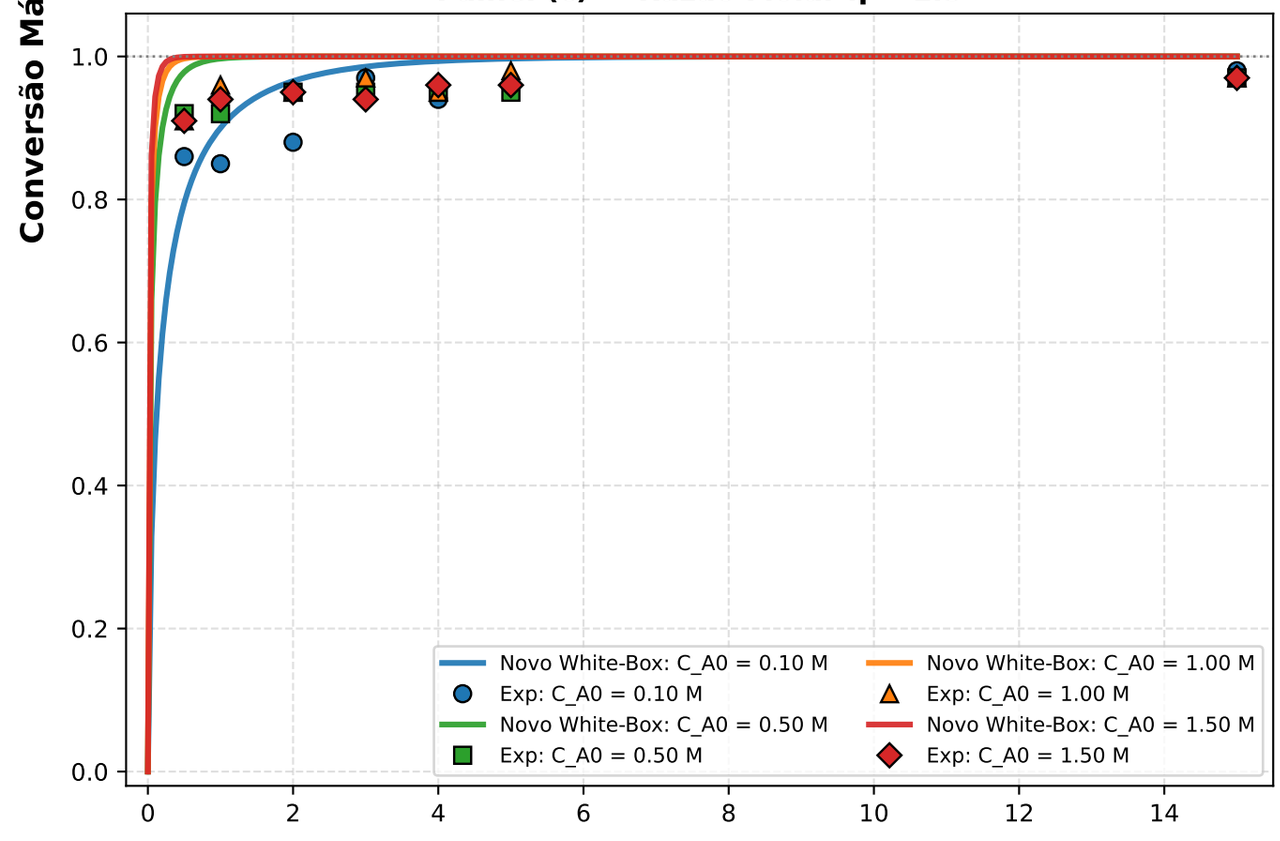
### Painel (a) — Razão Molar η = 0.5



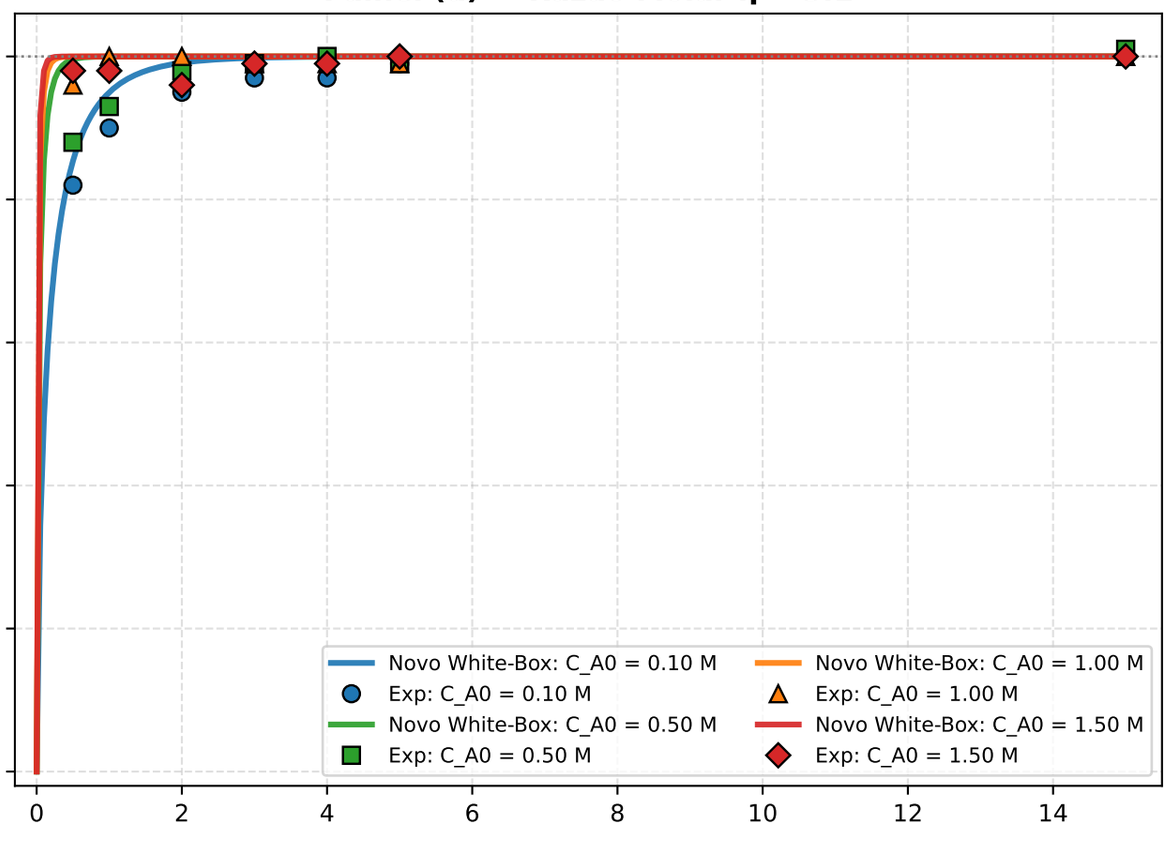

### Painel (b) — Razão Molar η = 1.0




### Painel (c) — Razão Molar η = 1.5




### Painel (d) — Razão Molar η = 3.1




**Conversão Mássica de Zinco, X_Zn (-)**

**Tempo de Reação, t (min)**